```c
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <limits.h>
#include <errno.h>

int main() {
    char path[PATH_MAX];
    char previous[PATH_MAX];
    char current[PATH_MAX];

    if (getcwd(current, sizeof(current)) == NULL) {
        perror("getcwd");
        return 1;
    }

    strcpy(previous, current);

    printf("Current directory: %s\n", current);
    printf("Enter directory path: ");

    if (fgets(path, sizeof(path), stdin) == NULL)
        return 1;

    path[strcspn(path, "\n")] = '\0';

    if (chdir(path) == 0) {
        printf("Directory changed successfully.\n");
        strcpy(previous, current);
    } else {
        perror("Invalid directory");
    }

    if (getcwd(current, sizeof(current)) != NULL)
        printf("Current directory: %s\n", current);
```

```c
        perror("Invalid directory");
    }

    if (getcwd(current, sizeof(current)) != NULL)
        printf("Current directory: %s\n", current);

    printf("Previous directory: %s\n", previous);

    return 0;
}
```

```
 builtin_shell.c     filecopy1                mmap_f
sravya@Sravya1955:~/os$ nano q9a.c
sravya@Sravya1955:~/os$ gcc q9a.c -o q9a
sravya@Sravya1955:~/os$ ./q9a
Current directory: /home/sravya/os
Enter directory path: /tmp
Directory changed successfully.
Current directory: /tmp
Previous directory: /home/sravya/os
```

```
 GNU nano 7.2

#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <stdlib.h>

void cmd_pwd(char *arg) {
    char path[1024];
    if (getcwd(path, sizeof(path)))
        printf("%s\n", path);
}

void cmd_cd(char *arg) {
    if (!arg || chdir(arg) != 0)
        perror("cd");
}

void cmd_exit(char *arg) {
    exit(0);
}

struct Command {
    char *name;
    void (*function)(char *);
};

struct Command commands[] = {
    {"pwd", cmd_pwd},
    {"cd", cmd_cd},
    {"exit", cmd_exit}
};

int main() {
    char line[256];
    char *name, *arg;

    while (1) {
```

```c
        printf("myshell> ");

        if (!fgets(line, sizeof(line), stdin))
            break;

        line[strcspn(line, "\n")] = '\0';

        name = strtok(line, " \t");
        arg = strtok(NULL, " \t");

        if (!name)
            continue;

        int found = 0;

        for (int i = 0; i < 3; i++) {
            if (strcmp(name, commands[i].name) == 0) {
                commands[i].function(arg);
                found = 1;
                break;
            }
        }

        if (!found)
            printf("Invalid command: %s\n", name);
    }

    return 0;
}
```

```
sravya@Sravya1955:~/os$ nano q9b.c
sravya@Sravya1955:~/os$ gcc q9b.c -o q9b
sravya@Sravya1955:~/os$ ./q9b
myshell> pwd
/home/sravya/os
myshell> cd
cd: Success
myshell> /tmp
Invalid command: /tmp
myshell> cd
cd: Success
myshell> hello
Invalid command: hello
myshell> exit
```